UNITY 6000.0.62F1 · COMPLETE OFFLINE DOCUMENTATION

# Serpent Clash Developer Guide

Install. Edit. Reskin. Build your release.

szekipapa77@gmail.com



CHAPTER 01

# Start here: Serpent Clash

Serpent Clash is a standalone editable Unity game template. Steer the serpent through the temple courtyard, eat gems to grow, swallow smaller rivals and avoid larger ones. Use the limited speed burst and follow the current Grow, Hunt, Survive or Crown match goal. Rival serpents are local computer opponents.

Includes 25 curated levels, source-backed generation, a game-specific visual editor and validation, 4 guided helpers, local saving and portrait/landscape scenes. Keep the required private game Kit and publisher-owned Szekipapa77Ads layer. Provider SDK binaries are excluded. There is no bundled HLK, purchases, multiplayer server or remote LiveOps backend. Import only one game Kit into a project because the Kit classes share names.

CHAPTER 02

# Requirements, installation and first play

Use Unity **6000.0.62f1**, URP **17.0.4** and Unity UI **2.0.0**. Install matching Android SDK/NDK/OpenJDK and WebGL modules in Unity Hub. Retain the supplied manifest and ProjectSettings.

1. Extract the standalone project to a short writable local path, then open it in the matching editor. Wait for package import and compilation.
2. For native unitypackage import, create a matching URP project and use Assets > Import Package > Custom Package. Import every game, Kit, Szekipapa77Ads and WebGL-template entry.
3. Run Szekipapa77 > Serpent Clash > Setup. It configures rendering and player settings; review its changes when integrating with existing work.
4. Active Input Handling must support the classic Input Manager: choose Old or Both. New Input System only is insufficient.
5. Open `Assets/Games/SerpentClash/Scenes/SerpentClash_Mobile.unity` and press Play. Follow the level card and first-level instructions.
6. Stop Play mode before editing level assets. Keep all .meta GUIDs. The game uses classic Unity UI Text; do not assume an extra TextMesh Pro import is required.

CHAPTER 03

# Project folders and exact source map

All game-relative paths below start at `Assets/Games/SerpentClash`. Runtime UI is built in code; editing an unrelated static Canvas does not reskin it. Scripts contains game rules and views; Editor contains authoring tools; Kit supplies UI/audio/save/platform support; Resources/serpentclash contains level JSON, art, music and sound; Settings owns URP assets; Scenes contains both entry scenes.

- `Scripts/ScData.cs`
- `Scripts/ScGen.cs`
- `Scripts/ScLevels.cs`
- `Scripts/ScMeshes.cs`
- `Scripts/ScSfx.cs`
- `Scripts/ScSim.cs`
- `Scripts/SerpentClashGame.cs`
- `Scripts/SerpentClashGame.UI.cs`
- `Scripts/SerpentClashGame.View.cs`

Main flow: SerpentClashGame.cs. UI and popup consumers: SerpentClashGame.UI.cs. Model/generator/loading: ScData.cs, ScGen.cs, ScLevels.cs, ScSim.cs. Authoring: Editor/SerpentClashLevelEditor.cs. Shared ad settings live outside the game at Assets/Szekipapa77Ads/Resources/AdsSettings.asset.

CHAPTER 04

# Gameplay, controls and progression

Steer the serpent through the temple courtyard, eat gems to grow, swallow smaller rivals and avoid larger ones. Use the limited speed burst and follow the current Grow, Hunt, Survive or Crown match goal. Rival serpents are local computer opponents.

The initial bank has 25 curated levels. Read the supplied loader/generator before changing continuation rules. Play the guided opening, ordinary progress, completion and retry paths; preserve saved indexes during a visual reskin. The Mobile scene is portrait and the WebGL-named scene is landscape, but the supplied Build menu selects portrait for both build targets. Select landscape explicitly in your Build Profile if desired. Test a near-startup scene replacement as well as a normal restart.

CHAPTER 05

# Tutorials and every helper

Fresh local progress triggers the first-level guidance. Newly unlocked helpers have introduction and guided-use flows. Reset only a test save when reviewing tutorials. A free guided first use can differ from ordinary charge consumption. Exact helper names/unlocks/charges: Scripts/SerpentClashGame.cs; help and effect descriptions: Scripts/SerpentClashGame.cs.

| Helper | Unlock | Free charges | Action and effect |
|---|---|---|---|
| Feast | 3 | 3 | A ring of 16 gems bursts out around you. Gobble them up! |
| Venom | 6 | 3 | For 6 seconds you can bite rivals up to 2 levels bigger than you. |
| Phantom | 10 | 2 | For 5 seconds you glide through serpents, walls and spikes. Nobody can bite you. |
| Stone Gaze | 15 | 2 | Every rival close to you turns to stone for 3 seconds. |

Activate the tool and perform its required target action: arming a tool alone does not demonstrate its effect. A successful use requires the stated preconditions from this game: show both activation and its actual effect. Exhausted charges may offer a rewarded ad; verify success/cancel/failure and grant a reward once only.

CHAPTER 06

# Visual level editor: worked editing session

Open **Szekipapa77 > Serpent Clash > Level Editor**. Exact implementation: `Editor/SerpentClashLevelEditor.cs`.

Duplicate an early entry in this game-specific visual editor, change its visible name and one rule, save, validate, then play it. Use the exact source-control table below for this game.

1. Back up Resources/serpentclash/Levels/levels.json.
2. Duplicate a simple early level and edit one small element.
3. Use the game-specific validation controls and resolve every reported issue.
4. Save explicitly; stop and restart Play mode, then use Play this level or navigate to the edited index.
5. Check portrait and landscape and rebuild demo outputs after accepting data changes.

The exact current authoring controls, including tooltips and generator switches, follow. These are game-specific source references, not a generic editor feature list.

| Source | Exact current code |
| --- | --- |
| `SerpentClashLevelEditor.cs:50` | `[MenuItem("Szekipapa77/Serpent Clash/Level Editor")]` |
| `SerpentClashLevelEditor.cs:53` | `[MenuItem("Szekipapa77/Serpent Clash/Validate All Levels")]` |
| `SerpentClashLevelEditor.cs:118` | `if (GUILayout.Button(_dirty ? "Save *" : "Save", EditorStyles.toolbarButton, GUILayout.Width(70))) Save();` |
| `SerpentClashLevelEditor.cs:119` | `if (GUILayout.Button("Reload", EditorStyles.toolbarButton, GUILayout.Width(60)) && (!_dirty \|\| EditorUtility.DisplayDialog("Reload", "Discard unsaved changes?", "Discard", "Cancel"))) Load();` |
| `SerpentClashLevelEditor.cs:122` | `if (GUILayout.Button("Validate", EditorStyles.toolbarButton, GUILayout.Width(70)) && Cur != null) ValidateOne(_sel);` |
| `SerpentClashLevelEditor.cs:123` | `if (GUILayout.Button("Validate All", EditorStyles.toolbarButton, GUILayout.Width(85))) ValidateAll();` |

| | |
|---|---|
| SerpentClashLevelEditor.cs:128 | if (GUILayout.Button(new GUIContent("Play this level", "Optional quick test: saves, opens the WebGL scene at this level and enters Play mode. Editing never needs it."), EditorStyles.toolbarButton, GUILayout.Width(100)) && Cur != null) Play(_sel + 1); |
| SerpentClashLevelEditor.cs:154 | if (GUILayout.Button("Add")) |
| SerpentClashLevelEditor.cs:159 | if (GUILayout.Button("Dup") && Cur != null) { var c = Cur.Clone(); c.name += " copy"; Insert(_sel + 1, c); } |
| SerpentClashLevelEditor.cs:160 | if (GUILayout.Button("Del") && Cur != null && EditorUtility.DisplayDialog("Delete", $"Delete level {_sel + 1} '{Cur.name}'?", "Delete", "Cancel")) |
| SerpentClashLevelEditor.cs:164 | if (GUILayout.Button("Up") && _sel > 0) Move(-1); |
| SerpentClashLevelEditor.cs:165 | if (GUILayout.Button("Down") && _sel < _file.levels.Length - 1) Move(1); |
| SerpentClashLevelEditor.cs:187 | EditorGUILayout.LabelField($"Level {_sel + 1}", EditorStyles.boldLabel); |
| SerpentClashLevelEditor.cs:189 | l.name = EditorGUILayout.TextField("Name", l.name); |
| SerpentClashLevelEditor.cs:190 | l.theme = EditorGUILayout.Popup("Theme", Mathf.Clamp(l.theme, 0, ThemeNames.Length - 1), ThemeNames, GUILayout.Width(260)); |
| SerpentClashLevelEditor.cs:192 | l.hint = EditorGUILayout.TextField("First-time hint", l.hint); |
| SerpentClashLevelEditor.cs:194 | l.difficulty = EditorGUILayout.Slider("Design difficulty", l.difficulty, 0, 1); |
| SerpentClashLevelEditor.cs:195 | l.breather = EditorGUILayout.ToggleLeft("Breather", l.breather, GUILayout.Width(80)); |
| SerpentClashLevelEditor.cs:196 | l.tut = EditorGUILayout.ToggleLeft("Tutorial", l.tut == 1, GUILayout.Width(80)) ? 1 : 0; |

| | |
|---|---|
| `SerpentClashLevelEditor.cs:199` | `_showSettings = EditorGUILayout.Foldout(_showSettings, "Match settings", true);` |
| `SerpentClashLevelEditor.cs:203` | `l.goal = EditorGUILayout.Popup("Goal", Mathf.Clamp(l.goal, 0, 3), GoalNames);` |
| `SerpentClashLevelEditor.cs:206` | `case Goal.Grow: l.target = EditorGUILayout.IntSlider("target serpent level", Mathf.Max(l.startLevel + 1, l.target), 2, 40); break;` |
| `SerpentClashLevelEditor.cs:207` | `case Goal.Hunt: l.target = EditorGUILayout.IntSlider("rivals to eat", Math.Max(1, l.target), 1, 8); break;` |
| `SerpentClashLevelEditor.cs:208` | `case Goal.Crown: l.target = EditorGUILayout.IntSlider("seconds on top", Math.Max(3, l.target), 3, 40); break;` |
| `SerpentClashLevelEditor.cs:209` | `default: EditorGUILayout.LabelField("survive the whole clock"); break;` |
| `SerpentClashLevelEditor.cs:213` | `l.time = EditorGUILayout.IntSlider("Clock (s)", l.time, 20, 600);` |
| `SerpentClashLevelEditor.cs:214` | `if (l.Goal != Goal.Survive && GUILayout.Button(new GUIContent("Calibrate", "Sets the clock from how long the auto player needs."), GUILayout.Width(80))) { ScGen.Calibrate(l, _sel + 3); _results.Remove(_sel); }` |
| `SerpentClashLevelEditor.cs:215` | `l.startLevel = EditorGUILayout.IntSlider("Player start level", Math.Max(1, l.startLevel), 1, 30);` |
| `SerpentClashLevelEditor.cs:218` | `l.bots = EditorGUILayout.IntSlider("Rivals", l.bots, 0, Sim.MaxSerp - 1);` |
| `SerpentClashLevelEditor.cs:219` | `l.aggro = EditorGUILayout.Slider("Aggression", l.aggro, 0, 1);` |
| `SerpentClashLevelEditor.cs:223` | `EditorGUILayout.MinMaxSlider($"Rival levels {(int)lo} - {(int)hi}", ref lo, ref hi, 1, 35);` |
| `SerpentClashLevelEditor.cs:225` | `l.leader = EditorGUILayout.IntSlider(new GUIContent("Crown leader level", "A rival who starts on top of the leaderboard (0 = none)."), l.leader, 0, 40);` |

| | |
|---|---|
| SerpentClashLevelEditor.cs:228 | `l.gems = EditorGUILayout.Slider("Gem amount", l.gems, 0.3f, 2.5f);` |
| SerpentClashLevelEditor.cs:229 | `l.pickups = EditorGUILayout.ToggleLeft("Speed and magnet runes", l.pickups, GUILayout.Width(170));` |
| SerpentClashLevelEditor.cs:230 | `l.ventPeriod = EditorGUILayout.Slider("Spike vent cycle (s)", l.ventPeriod, 2.5f, 10f);` |
| SerpentClashLevelEditor.cs:237 | `EditorGUILayout.HelpBox($"Auto player: {v} ({v.Seeds} seeds){(v.Solvable ? "" : "\nNOT reliably winnable - add time, fewer or smaller rivals, or more gems.")}", v.Solvable ? MessageType.Info : MessageType.Warning);` |
| SerpentClashLevelEditor.cs:247 | `_mirror = EditorGUILayout.ToggleLeft(new GUIContent("Mirror", "Also paints the point-symmetric tile, so both halves stay fair."), _mirror, GUILayout.Width(70));` |
| SerpentClashLevelEditor.cs:248 | `int w = EditorGUILayout.IntSlider("W", l.w, 8, 40, GUILayout.Width(200));` |
| SerpentClashLevelEditor.cs:249 | `int h = EditorGUILayout.IntSlider("H", l.h, 8, 40, GUILayout.Width(200));` |
| SerpentClashLevelEditor.cs:256 | `_stampWalls = EditorGUILayout.IntSlider("walls", _stampWalls, 0, 6, GUILayout.Width(190));` |
| SerpentClashLevelEditor.cs:257 | `_stampVents = EditorGUILayout.IntSlider("vents", _stampVents, 0, 8, GUILayout.Width(190));` |
| SerpentClashLevelEditor.cs:258 | `_stampReeds = EditorGUILayout.IntSlider("reeds", _stampReeds, 0, 4, GUILayout.Width(190));` |
| SerpentClashLevelEditor.cs:259 | `_stampPads = EditorGUILayout.IntSlider("pads", _stampPads, 0, 8, GUILayout.Width(190));` |
| SerpentClashLevelEditor.cs:260 | `if (GUILayout.Button("Stamp", GUILayout.Width(70)))` |
| SerpentClashLevelEditor.cs:266 | `EditorGUILayout.LabelField("Left-click/drag paints with the brush, right-click/drag paints moat. The gold dot is where you start (the flagstone nearest the middle).", EditorStyles.miniLabel);` |

| | |
|---|---|
| SerpentClashLevelEditor.cs:352 | `_showGen = EditorGUILayout.Foldout(_showGen, "Auto Generate (seeded, auto-player validated, same generator as the runtime)", true);` |
| SerpentClashLevelEditor.cs:355 | `_genCount = EditorGUILayout.IntField("Count", _genCount); _genStart = EditorGUILayout.IntField("Start at level #", _genStart);` |
| SerpentClashLevelEditor.cs:356 | `_genSeed = EditorGUILayout.IntField("Seed", _genSeed);` |
| SerpentClashLevelEditor.cs:359 | `_genFrom = EditorGUILayout.Slider("Start difficulty", _genFrom, 0, 1); _genTo = EditorGUILayout.Slider("End difficulty", _genTo, 0, 1);` |
| SerpentClashLevelEditor.cs:362 | `_genBreather = EditorGUILayout.IntSlider("Breather every", _genBreather, 0, 8);` |
| SerpentClashLevelEditor.cs:363 | `_genReplace = EditorGUILayout.ToggleLeft("Replace that range (else insert)", _genReplace);` |
| SerpentClashLevelEditor.cs:374 | `if (GUILayout.Button($"Generate {_genCount} level(s)", GUILayout.Height(26))) Generate();` |

CHAPTER 07

# Generator, insertion and replacement

Use this editor generation panel, choose a reproducible seed and small count, review start/index and replace behavior, then generate, save and reopen. The source excerpt below identifies exact append/replace semantics.

| Source | Exact current code |
| --- | --- |
| `SerpentClashLevelEditor.cs:48` | `bool _genReplace, _showGen = true, _showSettings = true;` |
| `SerpentClashLevelEditor.cs:363` | `_genReplace = EditorGUILayout.ToggleLeft("Replace that range (else insert)", _genReplace);` |
| `SerpentClashLevelEditor.cs:397` | `if (_genReplace) { int rm = Mathf.Min(_genCount, list.Count - at); if (rm > 0) list.RemoveRange(at, rm); }` |
| `SerpentClashLevelEditor.cs:398` | `list.InsertRange(at, made);` |

Worked extension: back up the bank, generate a small set at moderate difficulty using the exact per-game controls, validate them, then Save. Reopen the JSON and confirm the count changed. Do not assume every editor has the same insertion behavior: the actual implementation below distinguishes insert, append and replace. Runtime continuation does not automatically append new records to project JSON. Validation and bot scores help find problems but do not certify a pleasant human difficulty curve.

CHAPTER 08

# Actual level JSON and identifiers

Canonical asset: `Assets/Games/SerpentClash/Resources/serpentclash/Levels/levels.json`. Array order controls visible numbering. Preserve numeric/boolean types, root version fields where present and resource/save identifiers. The actual first record is shown below; use the editor for structured coordinates and block/word data.


```
{
  "name": "Sunstone Steps",
  "shape": "round",
  "goal": 0,
  "target": 4,
  "time": 60,
  "w": 15,
  "h": 15,
  "map": [
    "xxxxxx...xxxxxx",
    "xxxx.......xxxx",
    "xx...........xx",
    "xx...........xx",
    "x.............x",
    "x.............x",
    ".............",
    ".............",
    ".............",
    "x.............x",
    "x.............x",
    "xx...........xx",
    "xx...........xx",
    "xxxx.......xxxx",
    "xxxxxx...xxxxxx"
  ],
  "bots": 4,
  "botMin": 1,
  "botMax": 1,
  "aggro": 0,
  "startLevel": 1,
  "gems": 1.3,
  "pickups": false,
  "leader": 0,
  "ventPeriod": 5.35,
  "theme": 0,
  "breather": false,
  "difficulty": 0.1,
  "hint": "",
  "tut": 1
}
```


Root keys: levels. Inspect ScData.cs, ScGen.cs, ScLevels.cs, ScSim.cs for field semantics and validation. Keep visible names/hints distinct from technical keys. The game-specific board, wave, card, opponent or question records are gameplay data; UI translation is separate.

CHAPTER 09

# Reskin recipe: interface colours

Stop Play mode and open `Scripts/SerpentClashGame.UI.cs`. Find `Ink`, currently `#2E2418`. Change that hex literal to `#42C9B0`, preserve its symbol/type, compile and reopen the scene. Review HUD, menus, helper cards, settings/results and both orientations. Art sprites and procedural fallback shapes can have additional colours; edit them deliberately.

Actual consumers of this symbol are listed first, then all current palette/hex assignments. Editor-only palettes do not control the runtime. Preserve state/goal contrast and rebuild after accepting a change.

| Source | Exact current code |
| --- | --- |
| `SerpentClashGame.UI.cs:104 / void BuildHud()` | `_goalText = HzUI.Label("GoalT", _goalPanel, "", 42, Ink, TextAnchor.MiddleLeft);` |
| `SerpentClashGame.UI.cs:226 / void BuildHud()` | `_tipText = HzUI.Label("T", _tipPanel, "", 40, Ink); _tipText.rectTransform.Stretch(20);` |
| `SerpentClashGame.UI.cs:310 / void BuildHud()` | `v.Caption = HzUI.Label("Cap", v.Root, "", 30, Ink);` |
| `SerpentClashGame.UI.cs:330 / void RefreshBoosters()` | `v.Caption.color = un ? Ink : HzUI.Hex("#8C7A5E");` |

| Source | Exact current code |
| --- | --- |
| `SerpentClashGame.cs:29 / class/static declarations` | `static readonly string[] BoostColor = { "#FF9F43", "#3FCB6A", "#5AA9FF", "#A98BFF" };` |
| `SerpentClashGame.cs:139 / IEnumerator Intro(string key, string title, string msg, string art, string icon = null)` | `TemplePopup(title, msg, HzUI.Hex("#1F8A70"), Art(art), icon, ("Got it!", HzUI.Hex("#F2A93B"), () => closed = true));` |
| `SerpentClashGame.cs:287 / void ProcessEvents()` | `FloatText(PlayerPos + Vector3.up * 2.2f, "Level " + e.Index + "!", HzUI.Hex("#F2C14E"));` |
| `SerpentClashGame.cs:294 / void ProcessEvents()` | `if (e.A == 0) { HzAudio.Play(Sfx.LevelDown, 0.6f); FloatText(PlayerPos + Vector3.up * 2f, "Level " + e.Index, HzUI.Hex("#FF8A7A")); UpdateGoalUi(false); }` |
| `SerpentClashGame.cs:301 / void ProcessEvents()` | `FloatText(at + Vector3.up * 1.5f, "-" + e.Index, HzUI.Hex("#FF6B57"));` |
| `SerpentClashGame.cs:302 / void ProcessEvents()` | `if (e.A > 0) Feed($"{_sim.Serpents[e.A].Name} bit your tail!", HzUI.Hex("#FF9C8A"));` |

| | |
|---|---|
| `SerpentClashGame.cs:309 / void ProcessEvents()` | `FloatText(at + Vector3.up * 1.6f, "Snap! -" + e.Index, HzUI.Hex("#7CF0B4"));` |
| `SerpentClashGame.cs:323 / void ProcessEvents()` | `Feed($"{killer} swallowed you!", HzUI.Hex("#FF8A7A"));` |
| `SerpentClashGame.cs:329 / void ProcessEvents()` | `FloatText(at + Vector3.up * 2.2f, "GULP!", HzUI.Hex("#F2C14E"));` |
| `SerpentClashGame.cs:330 / void ProcessEvents()` | `Feed($"You swallowed {v.Name}!", HzUI.Hex("#F2C14E"));` |
| `SerpentClashGame.cs:354 / void ProcessEvents()` | `if (e.B == 0) { HzAudio.Play(Sfx.VentCut, 0.85f); ShakeCamera(0.15f, 0.25f); FloatText(at + Vector3.up * 1.5f, "Ouch! -" + Mathf.Max(1, e.Index), HzUI.Hex("#FFB347")); }` |
| `SerpentClashGame.cs:400 / void ProcessEvents()` | `if (crown && _sim.Time > 1) { HzAudio.Play(Sfx.Crown, 0.7f); Feed("You hold the sun crown!", HzUI.Hex("#F2C14E")); }` |
| `SerpentClashGame.cs:431 / IEnumerator WinFlow()` | `var pop = TemplePopup("Victory!", msg, HzUI.Hex("#F2A93B"), Art("win_badge"), null,` |
| `SerpentClashGame.cs:432 / IEnumerator WinFlow()` | `("Next Level", HzUI.Hex("#1F8A70"), () => HzPlatform.Interstitial(GameId, () => LoadLevel(Level))));` |
| `SerpentClashGame.cs:451 / IEnumerator LoseFlow()` | `TemplePopup(title, msg + (reason == "time" ? "\nTake 30 more seconds?" : "\nCome back with most of your size?"), HzUI.Hex("#B5523B"), Art("lose_badge"), null,` |
| `SerpentClashGame.cs:452 / IEnumerator LoseFlow()` | `(label, HzUI.Hex("#1F8A70"), () => HzPlatform.Rewarded(GameId, "rescue", ok => { if (ok) Rescue(); else LoadLevel(Level, true); })),` |
| `SerpentClashGame.cs:453 / IEnumerator LoseFlow()` | `("Retry", HzUI.Hex("#F2A93B"), () => LoadLevel(Level, true)));` |
| `SerpentClashGame.cs:456 / IEnumerator LoseFlow()` | `TemplePopup(title, msg, HzUI.Hex("#B5523B"), Art("lose_badge"), null, ("Retry", HzUI.Hex("#F2A93B"), () => LoadLevel(Level, true)));` |

| | |
|---|---|
| SerpentClashGame.cs:504 / void OnBooster(int b) | ("+1 (ad)", HzUI.Hex("#1F8A70"), () => HzPlatform.Rewarded(GameId, "booster_" + BoostShort[b].ToLowerInvariant(), ok => { if (ok) { SetCharges(b, Charges(b) + 1); Toast("+1 " + BoostName[b]); HzAudio.Play(Sfx.Coin, 0.7f); } })), |
| SerpentClashGame.cs:505 / void OnBooster(int b) | ("No thanks", HzUI.Hex("#B9A98C"), null)); |
| SerpentClashGame.cs:514 / void OnBooster(int b) | case BFeast: _sim.Feast(); HzAudio.Play(Sfx.Feast, 0.85f); SpawnSparks(pos + Vector3.up, new Color(2f, 1.5f, 0.6f), 16, 1.2f); FloatText(pos + Vector3.up * 2f, "FEAST!", HzUI.Hex("#FF9F43")); break; |
| SerpentClashGame.cs:515 / void OnBooster(int b) | case BVenom: _sim.Venom(); HzAudio.Play(Sfx.Venom, 0.85f); SpawnBurst(pos + Vector3.up * 0.05f, new Color(0.6f, 2f, 0.8f), 4f); FloatText(pos + Vector3.up * 2f, "VENOM!", HzUI.Hex("#3FCB6A")); break; |
| SerpentClashGame.cs:516 / void OnBooster(int b) | case BPhantom: _sim.Phantom(); HzAudio.Play(Sfx.Phantom, 0.85f); SpawnBurst(pos + Vector3.up * 0.05f, new Color(0.7f, 1.3f, 2.2f), 4f); FloatText(pos + Vector3.up * 2f, "PHANTOM!", HzUI.Hex("#8CC8FF")); break; |
| SerpentClashGame.cs:522 / void OnBooster(int b) | FloatText(pos + Vector3.up * 2f, n > 0 ? "STONE GAZE x" + n : "STONE GAZE", HzUI.Hex("#C9B8FF")); |
| SerpentClashGame.UI.cs:13 / class/static declarations | static readonly Color Ink = HzUI.Hex("#2E2418"); |
| SerpentClashGame.UI.cs:14 / class/static declarations | static readonly Color Jade = HzUI.Hex("#1F8A70"); |
| SerpentClashGame.UI.cs:15 / class/static declarations | static readonly Color DeepTeal = HzUI.Hex("#0E3B43"); |
| SerpentClashGame.UI.cs:16 / class/static declarations | static readonly Color Gold = HzUI.Hex("#F2A93B"); |
| SerpentClashGame.UI.cs:17 / class/static declarations | static readonly Color Ivory = HzUI.Hex("#FFF6E3"); |
| SerpentClashGame.UI.cs:18 / class/static declarations | static readonly Color TagSmall = HzUI.Hex("#8CF5B8"), TagBig = HzUI.Hex("#FF9C8A"), TagSame = HzUI.Hex("#FFF3D6"); |

| | |
|---|---|
| SerpentClashGame.UI.cs:84 / void BuildHud() | _levelSub = HzUI.Label("Name", lp.transform, _def.name + (_def.breather ? " (easy)" : ""), 30, HzUI.Hex("#E2FFF4"), TextAnchor.MiddleCenter, false); |
| SerpentClashGame.UI.cs:108 / void BuildHud() | var barBg = HzUI.Panel("BarBg", _goalPanel, HzUI.Hex("#EADFC6"), 20); |
| SerpentClashGame.UI.cs:134 / void BuildHud() | var cap = HzUI.Label("Cap", _serpCard, "LEVEL", 24, HzUI.Hex("#CFEFE4"), TextAnchor.MiddleLeft, false); cap.fontStyle = FontStyle.Bold; |
| SerpentClashGame.UI.cs:136 / void BuildHud() | _xpText = HzUI.Label("Xp", _serpCard, "", 24, HzUI.Hex("#F7CD52"), TextAnchor.MiddleRight, false); _xpText.fontStyle = FontStyle.Bold; |
| SerpentClashGame.UI.cs:140 / void BuildHud() | _xpFill = HzUI.Panel("XpFill", xpBg.transform, HzUI.Hex("#F7CD52"), 12); _xpFill.raycastTarget = false; |
| SerpentClashGame.UI.cs:196 / void BuildHud() | var sp = HzUI.Panel("Boost", bar, HzUI.Hex("#FF8A3D"), 100); |
| SerpentClashGame.UI.cs:249 / void BuildHud() | _energyFill = HzUI.Panel("Fill", _energy, HzUI.Hex("#FFB347"), 8); _energyFill.raycastTarget = false; |
| SerpentClashGame.UI.cs:305 / void BuildHud() | var bd = HzUI.Panel("Badge", v.Root, HzUI.Hex("#E2453C"), 30); bd.raycastTarget = false; |
| SerpentClashGame.UI.cs:322 / void RefreshBoosters() | v.Inner.color = un ? HzUI.Hex(BoostColor[b]) : HzUI.Hex("#CDBFA4"); |
| SerpentClashGame.UI.cs:328 / void RefreshBoosters() | v.Badge.GetComponent<Image>().color = c > 0 ? HzUI.Hex("#E2453C") : Jade; |
| SerpentClashGame.UI.cs:330 / void RefreshBoosters() | v.Caption.color = un ? Ink : HzUI.Hex("#8C7A5E"); |
| SerpentClashGame.UI.cs:331 / void RefreshBoosters() | v.Ring.color = _guide == b ? HzUI.Hex("#FFE066") : new Color(1, 1, 1, 0); |
| SerpentClashGame.UI.cs:347 / void UpdateHud(float dt) | _timerBg.color = warn ? HzUI.Hex("#E2453C") : _def.Goal == Goal.Survive ? HzUI.Hex("#2FA36B") : Jade; |
| SerpentClashGame.UI.cs:392 / void UpdateHud(float dt) | tag.color = mode == 3 ? HzUI.Hex("#FFE08A") : mode == 0 ? TagSmall : mode == 1 ? TagBig : mode == 4 ? HzUI.Hex("#D8D3C8") : TagSame; |

| | |
|---|---|
| `SerpentClashGame.UI.cs:407 / void UpdateHud(float dt)` | `_energyFill.color = p.Boosting ? HzUI.Hex("#FFB347") : HzUI.Hex("#9FD8C8");` |
| `SerpentClashGame.UI.cs:455 / void RefreshBoard()` | `_boardName[r].color = idx == 0 ? HzUI.Hex("#FFD166") : Color.white;` |
| `SerpentClashGame.UI.cs:551 / void LevelBanner()` | `var n = HzUI.Label("N", bg.transform, _def.name + " - " + GoalShort(), 44, HzUI.Hex("#FFE2A8")); n.rectTransform.Place(new Vector2(0.5f, 0), new Vector2(820, 70), new Vector2(0, 22), new Vector2(0.5f, 0));` |
| `SerpentClashGame.UI.cs:616 / void CompleteGuide(int b)` | `Toast("Nice! " + BoostName[b] + " is yours", HzUI.Hex("#FFE2A8"), 1.3f, IsPortrait ? 0.62f : 0.5f);` |
| `SerpentClashGame.UI.cs:659 / HzPopup TemplePopup(string title, string message, Color accent, Sprite art, string icon, params (string label, Color color, Action action)[] buttons)` | `var card = HzUI.Panel("Card", dim.transform, HzUI.Hex("#E6CFA0"), 56);` |
| `SerpentClashGame.UI.cs:662 / HzPopup TemplePopup(string title, string message, Color accent, Sprite art, string icon, params (string label, Color color, Action action)[] buttons)` | `card.AddOutline(HzUI.Hex("#B98A3E"), 4);` |
| `SerpentClashGame.UI.cs:710 / HzPopup TemplePopup(string title, string message, Color accent, Sprite art, string icon, params (string label, Color color, Action action)[] buttons)` | `p.Message = HzUI.Label("Msg", card.transform, message, 42, HzUI.Hex("#3D3020"), TextAnchor.MiddleCenter, false);` |
| `SerpentClashGame.UI.cs:741 / void AddStars(HzPopup pop, int stars)` | `var st = HzIcons.Icon(row, HzIcons.Star, s == 1 ? 120 : 96, on ? HzUI.Hex("#FFC93C") : new Color(0.8f, 0.75f, 0.65f, 0.9f));` |
| `SerpentClashGame.UI.cs:742 / void AddStars(HzPopup pop, int stars)` | `st.AddOutline(HzUI.Hex("#A8681E"), on ? 4 : 0.01f);` |
| `SerpentClashGame.UI.cs:761 / void ShowSettingsPopup()` | `(SoundLabel(), HzUI.Hex("#3E8ED0"), null),` |
| `SerpentClashGame.UI.cs:762 / void ShowSettingsPopup()` | `(MusicLabel(), HzUI.Hex("#8E6BD8"), null),` |
| `SerpentClashGame.UI.cs:763 / void ShowSettingsPopup()` | `("Restart Level", HzUI.Hex("#F2A93B"), () => LoadLevel(Level, true)),` |

| | |
|---|---|
| `SerpentClashGame.View.cs:17 / class/static declarations` | `static readonly string[] SkinA = { "#19B48A", "#8B5CF6", "#FF6B57", "#4FB6FF", "#FFC93C", "#FF7AD9", "#7BD66F", "#FF9F45", "#3FD3C6", "#C77DFF", "#F25F5C", "#5C7CFA" };` |
| `SerpentClashGame.View.cs:18 / class/static declarations` | `static readonly string[] SkinB = { "#F7CD52", "#FFD166", "#FFE6C7", "#1F4FA8", "#E46A00", "#7A1F5C", "#1E6B3A", "#FFF1C1", "#0B6E69", "#FFE3F3", "#2B2D42", "#BFE3FF" };` |
| `SerpentClashGame.View.cs:31 / class/static declarations` | `new Theme { StoneA = "#D9BC8A", StoneB = "#CDAE7B", StoneC = "#E0C79A", Grout = "#B8956A", Curb = "#C9A46A", Wall = "#C8A77A", Cap = "#E9B949", Rune = "#3FE0C5", Water = "#1FA7B5", Moss = "#9DBA6A", Reed = "#5E9E45", Bronze = "#A9743A", Fog = "#7FC9C6", AmbSky = "#BFE6F0", AmbEq = "#F2D9B0", AmbGround = "#7A5E44", Sun = "#FFE6C2", WaterGlow = 0.18f, SunIntensity = 1.0f, FlameGlow = 1f },` |
| `SerpentClashGame.View.cs:33 / class/static declarations` | `new Theme { StoneA = "#A9BFA2", StoneB = "#98B091", StoneC = "#B7CBB0", Grout = "#6F8A6E", Curb = "#7FA07A", Wall = "#7E9A80", Cap = "#E0B547", Rune = "#7CFFB2", Water = "#1D5FA8", Moss = "#6E9E52", Reed = "#3F8A3C", Bronze = "#9C6B33", Fog = "#5E8FA6", AmbSky = "#B9D9E8", AmbEq = "#CFE3C4", AmbGround = "#3E5240", Sun = "#FFF0D2", WaterGlow = 0.12f, SunIntensity = 1.1f, FlameGlow = 1f },` |
| `SerpentClashGame.View.cs:35 / class/static declarations` | `new Theme { StoneA = "#6B5B57", StoneB = "#5E504D", StoneC = "#786663", Grout = "#3E3232", Curb = "#8C5A3C", Wall = "#4F4341", Cap = "#E8A23A", Rune = "#FFB347", Water = "#B83A12", Moss = "#7A6A3A", Reed = "#8A7A3A", Bronze = "#B07A3C", Fog = "#5A2E24", AmbSky = "#E3A07A", AmbEq = "#A8705A", AmbGround = "#3A2420", Sun = "#FFD2A8", WaterGlow = 0.8f, SunIntensity = 1.0f, FlameGlow = 1.3f, Lava = true },` |
| `SerpentClashGame.View.cs:37 / class/static declarations` | `new Theme { StoneA = "#A7B3C9", StoneB = "#96A3BB", StoneC = "#B6C1D4", Grout = "#6A7592", Curb = "#8C9AB8", Wall = "#7F8BA6", Cap = "#D9C27A", Rune = "#8FD8FF", Water = "#2A4C86", Moss = "#6F8F86", Reed = "#4F7F72", Bronze = "#8F7A55", Fog = "#26365E", AmbSky = "#5F72A8", AmbEq = "#6C7AA6", AmbGround = "#262B42", Sun = "#C8D8FF", WaterGlow = 0.22f, SunIntensity = 0.75f, FlameGlow = 1.6f, Night = true },` |

| | |
|---|---|
| SerpentClashGame.View.cs:39 / class/static declarations | new Theme { StoneA = "#E6C1C7", StoneB = "#DBB2B9", StoneC = "#EDCDD2", Grout = "#C29AA4", Curb = "#D7A1AE", Wall = "#D9B2B8", Cap = "#E9B949", Rune = "#FF8FC7", Water = "#3FB7A8", Moss = "#8FBF7A", Reed = "#5FA052", Bronze = "#B07A4A", Fog = "#9ED6CC", AmbSky = "#D8EEF0", AmbEq = "#F6DCE0", AmbGround = "#7A5A62", Sun = "#FFEBDD", WaterGlow = 0.15f, SunIntensity = 1.0f, FlameGlow = 1f }, |
| SerpentClashGame.View.cs:97 / static Color FoodColor(int kind, int skin) => kind == 3 ? SkinColor(skin, 0) : HzUI.Hex(kind == 0 ? "#19C9B6" : kind == 1 ? "#E8336B" : "#FFB21F"); | static Color FoodColor(int kind, int skin) => kind == 3 ? SkinColor(skin, 0) : HzUI.Hex(kind == 0 ? "#19C9B6" : kind == 1 ? "#E8336B" : "#FFB21F"); |
| SerpentClashGame.View.cs:98 / static Color PickColor(int kind) => kind == (int)Pick.Speed ? HzUI.Hex("#4FD2FF") : HzUI.Hex("#B07CFF"); | static Color PickColor(int kind) => kind == (int)Pick.Speed ? HzUI.Hex("#4FD2FF") : HzUI.Hex("#B07CFF"); |
| SerpentClashGame.View.cs:147 / void SetupLook() | _stoneMat = Lit("#9A968C", 0.15f); |
| SerpentClashGame.View.cs:149 / void SetupLook() | _eyeMat = Lit("#FFFDF5", 0.9f); |
| SerpentClashGame.View.cs:150 / void SetupLook() | _pupilMat = Lit("#1A1420", 0.95f); |
| SerpentClashGame.View.cs:151 / void SetupLook() | _tongueMat = Lit("#E2384D", 0.6f); |
| SerpentClashGame.View.cs:152 / void SetupLook() | _haloMat = Emit(HzUI.Hex("#FFC93C"), 0.9f, HzUI.Hex("#FFB020") * 1.4f); _haloMat.SetFloat("_Metallic", 0.5f); |
| SerpentClashGame.View.cs:153 / void SetupLook() | _ringPlayer = Unlit(HzUI.Hex("#FFD166") * 1.3f); |
| SerpentClashGame.View.cs:160 / void SetupLook() | _padMat = Unlit(HzUI.Hex("#5FF2E0") * 1.6f); |
| SerpentClashGame.View.cs:235 / void BuildWorld() | _fill.color = th.Lava ? HzUI.Hex("#FF8A5C") : th.Night ? HzUI.Hex("#7F9BFF") : HzUI.Hex("#BFD6FF"); |
| SerpentClashGame.View.cs:244 / void BuildWorld() | _ventMat = Lit("#5B4632", 0.4f); |
| SerpentClashGame.View.cs:245 / void BuildWorld() | _ventHot = Emit(HzUI.Hex("#7A3A1A"), 0.4f, HzUI.Hex("#FF6A1F") * 1.6f); |

| | |
|---|---|
| `SerpentClashGame.View.cs:372 / void BuildDecor(Material stone, Material cap, Material bronze, System.Random rng)` | `if (lilies.Count > 0) MeshObj(_theme.Lava ? "Rocks" : "Lilies", _world, ScMeshes.Combine("lilies", lilies), _theme.Lava ? Lit("#2E2422", 0.2f) : Lit("#4E9A4A", 0.55f), false, true);` |
| `SerpentClashGame.View.cs:382 / void BuildDecor(Material stone, Material cap, Material bronze, System.Random rng)` | `l.type = LightType.Point; l.color = HzUI.Hex("#FFB062"); l.range = 11f; l.intensity = 1.4f * _theme.FlameGlow; l.shadows = LightShadows.None;` |

CHAPTER 10

# Reskin recipe: visible menu and HUD text

Open `Scripts/SerpentClashGame.UI.cs`, find the displayed literal `"Settings"`, and replace its displayed value with `"Options"`. Keep hierarchy names and technical IDs unchanged. In HzUI.Label/Button calls the first string may be an object name; change the displayed text argument. Read the method references below to distinguish construction from later refresh assignments.

Preserve interpolation braces, invariant number formatting, ON/OFF updates and line breaks. Update both initial and refresh paths, portrait and landscape. Exact helper names/unlocks/charges: Scripts/SerpentClashGame.cs; help and effect descriptions: Scripts/SerpentClashGame.cs. Level names/hints remain separate JSON content. If a title appears in art, Title property or another screen, update those visible uses deliberately without renaming GameId, namespace, resources or save keys.

Hard-coded game strings do not automatically use shared localization. To translate the interface, route every display assignment through your translation table. Dictionaries, word banks and theme pools are independent gameplay assets and need separate validation. Test help, settings, results, tutorial and helper introductions after a fresh launch.

| Source | Exact current code |
| --- | --- |
| `ScSim.cs:58 / public static float FoodValue(int kind) => kind == 0 ? 1f : kind == 1 ? 3f : 8f;` | `public string Name = "";` |
| `ScSim.cs:213 / public void Layout()` | `Player.Name = "You";` |
| `SerpentClashGame.cs:25 / class/static declarations` | `static readonly string[] BoostName = { "Feast", "Venom", "Phantom", "Stone Gaze" };` |
| `SerpentClashGame.cs:139 / IEnumerator Intro(string key, string title, string msg, string art, string icon = null)` | `TemplePopup(title, msg, HzUI.Hex("#1F8A70"), Art(art), icon, ("Got it!", HzUI.Hex("#F2A93B"), () => closed = true));` |
| `SerpentClashGame.cs:431 / IEnumerator WinFlow()` | `var pop = TemplePopup("Victory!", msg, HzUI.Hex("#F2A93B"), Art("win_badge"), null,` |
| `SerpentClashGame.cs:451 / IEnumerator LoseFlow()` | `TemplePopup(title, msg + (reason == "time" ? "\nTake 30 more seconds?" : "\nCome back with most of your size?"), HzUI.Hex("#B5523B"), Art("lose_badge"), null,` |
| `SerpentClashGame.cs:456 / IEnumerator LoseFlow()` | `TemplePopup(title, msg, HzUI.Hex("#B5523B"), Art("lose_badge"), null, ("Retry", HzUI.Hex("#F2A93B"), () => LoadLevel(Level, true)));` |

| | |
|---|---|
| `SerpentClashGame.cs:497 / void OnBooster(int b)` | `if (!Unlocked(b)) { Toast("Unlocks at level " + BoostUnlock[b]); HzAudio.Play(Sfx.Wrong, 0.5f); if (_boost[b] != null) HzTween.Shake(_boost[b].Root, 10, 0.3f); return; }` |
| `SerpentClashGame.cs:503 / void OnBooster(int b)` | `TemplePopup("Need " + BoostName[b] + "?", "Watch a short ad to get one more.", HzUI.Hex(BoostColor[b]), Art(BoostArt[b]), BoostIcon[b],` |
| `SerpentClashGame.cs:504 / void OnBooster(int b)` | `("+1 (ad)", HzUI.Hex("#1F8A70"), () => HzPlatform.Rewarded(GameId, "booster_" + BoostShort[b].ToLowerInvariant(), ok => { if (ok) { SetCharges(b, Charges(b) + 1); Toast("+1 " + BoostName[b]); HzAudio.Play(Sfx.Coin, 0.7f); } })),` |
| `SerpentClashGame.cs:579 / public override string DebugCommand(string cmd)` | `case "settings": ShowSettingsPopup(); return "ok";` |
| `SerpentClashGame.UI.cs:81 / void BuildHud()` | `_levelText = HzUI.Label("Lvl", lp.transform, "Level " + Level, 54, Color.white);` |
| `SerpentClashGame.UI.cs:84 / void BuildHud()` | `_levelSub = HzUI.Label("Name", lp.transform, _def.name + (_def.breather ? " (easy)" : ""), 30, HzUI.Hex("#E2FFF4"), TextAnchor.MiddleCenter, false);` |
| `SerpentClashGame.UI.cs:104 / void BuildHud()` | `_goalText = HzUI.Label("GoalT", _goalPanel, "", 42, Ink, TextAnchor.MiddleLeft);` |
| `SerpentClashGame.UI.cs:106 / void BuildHud()` | `_goalSub = HzUI.Label("GoalS", _goalPanel, "", 34, Jade, TextAnchor.MiddleRight);` |
| `SerpentClashGame.UI.cs:119 / void BuildHud()` | `_timerText = HzUI.Label("T", _timerBg.transform, "", 46, Color.white); _timerText.rectTransform.Place(new Vector2(1, 0.5f), new Vector2(118, 80), new Vector2(-64, 0));` |
| `SerpentClashGame.UI.cs:132 / void BuildHud()` | `_lvText = HzUI.Label("N", badge.transform, "1", 44, Color.white); _lvText.rectTransform.Stretch();` |
| `SerpentClashGame.UI.cs:134 / void BuildHud()` | `var cap = HzUI.Label("Cap", _serpCard, "LEVEL", 24, HzUI.Hex("#CFEFE4"), TextAnchor.MiddleLeft, false); cap.fontStyle = FontStyle.Bold;` |

| | |
|---|---|
| `SerpentClashGame.UI.cs:136 / void BuildHud()` | `_xpText = HzUI.Label("Xp", _serpCard, "", 24, HzUI.Hex("#F7CD52"), TextAnchor.MiddleRight, false); _xpText.fontStyle = FontStyle.Bold;` |
| `SerpentClashGame.UI.cs:157 / void BuildHud()` | `_boardRank[r] = HzUI.Label("Rank", row.transform, (r + 1).ToString(), 30, new Color(1, 1, 1, 0.7f));` |
| `SerpentClashGame.UI.cs:161 / void BuildHud()` | `_boardName[r] = HzUI.Label("Name", row.transform, "", 30, Color.white, TextAnchor.MiddleLeft, false);` |
| `SerpentClashGame.UI.cs:164 / void BuildHud()` | `_boardVal[r] = HzUI.Label("Val", row.transform, "", 30, Color.white, TextAnchor.MiddleRight);` |
| `SerpentClashGame.UI.cs:205 / void BuildHud()` | `var st = HzUI.Label("Cap", sp.transform, "Boost", 38, Color.white); st.rectTransform.Place(new Vector2(0.5f, 0), new Vector2(200, 50), new Vector2(0, 38));` |
| `SerpentClashGame.UI.cs:214 / void BuildHud()` | `var hint = HzUI.Label("T", hp.transform, "Hold click or Space to boost - keys 1-4 use boosters", 28, Color.white, TextAnchor.MiddleCenter, false);` |
| `SerpentClashGame.UI.cs:226 / void BuildHud()` | `_tipText = HzUI.Label("T", _tipPanel, "", 40, Ink); _tipText.rectTransform.Stretch(20);` |
| `SerpentClashGame.UI.cs:233 / void BuildHud()` | `_startText = HzUI.Label("T", _startPrompt, portrait ? "Drag anywhere to start!" : "Click to start!", 54, Color.white); _startText.rectTransform.Stretch(8);` |
| `SerpentClashGame.UI.cs:257 / void BuildHud()` | `var tag = HzUI.Label("Tag", _labels, "", 30, Color.white, TextAnchor.MiddleCenter, false);` |
| `SerpentClashGame.UI.cs:282 / void BuildHud()` | `v.Btn = HzUI.Button("Booster" + b, parent, null, Ivory, Color.white, () => OnBooster(id), 40, (int)(size / 2));` |
| `SerpentClashGame.UI.cs:308 / void BuildHud()` | `v.BadgeText = HzUI.Label("N", bd.transform, "", 34, Color.white); v.BadgeText.rectTransform.Stretch();` |
| `SerpentClashGame.UI.cs:310 / void BuildHud()` | `v.Caption = HzUI.Label("Cap", v.Root, "", 30, Ink);` |
| `SerpentClashGame.UI.cs:327 / void RefreshBoosters()` | `v.BadgeText.text = c > 0 ? c.ToString() : "+";` |

| | |
|---|---|
| SerpentClashGame.UI.cs:329 / void RefreshBoosters() | v.Caption.text = un ? (IsPortrait ? BoostShort[b] : (b + 1) + " " + BoostShort[b]) : "Lv " + BoostUnlock[b]; |
| SerpentClashGame.UI.cs:345 / void UpdateHud(float dt) | _timerText.text = $"{sec / 60}:{sec % 60:00}"; |
| SerpentClashGame.UI.cs:368 / void UpdateHud(float dt) | _xpText.text = Num(Mathf.FloorToInt(Mathf.Max(0, hi - p.Xp))) + " to Lv " + Num(p.Level + 1); |
| SerpentClashGame.UI.cs:391 / void UpdateHud(float dt) | tag.text = (s.IsPlayer ? "You" : s.Name) + " " + Num(s.Level); |
| SerpentClashGame.UI.cs:445 / void RefreshBoard() | if (idx >= n) { _boardName[r].text = ""; _boardVal[r].text = ""; _boardDot[r].color = new Color(1, 1, 1, 0); _boardBg[r].color = new Color(1, 1, 1, 0); continue; } |
| SerpentClashGame.UI.cs:509 / void Feed(string text, Color c) | var l = HzUI.Label("T", bg.transform, text, 32, c, TextAnchor.MiddleCenter, false); l.fontStyle = FontStyle.Bold; |
| SerpentClashGame.UI.cs:522 / void FloatText(Vector3 world, string text, Color c) | var l = HzUI.Label("Float", _labels, text, 56, Color.white); |
| SerpentClashGame.UI.cs:549 / void LevelBanner() | var t = HzUI.Label("L", bg.transform, "Level " + Level, 92, Color.white); t.rectTransform.Place(new Vector2(0.5f, 1), new Vector2(820, 110), new Vector2(0, -14), new Vector2(0.5f, 1)); |
| SerpentClashGame.UI.cs:551 / void LevelBanner() | var n = HzUI.Label("N", bg.transform, _def.name + " - " + GoalShort(), 44, HzUI.Hex("#FFE2A8")); n.rectTransform.Place(new Vector2(0.5f, 0), new Vector2(820, 70), new Vector2(0, 22), new Vector2(0.5f, 0)); |
| SerpentClashGame.UI.cs:603 / void RefreshGuide() | ShowTip($"New: {BoostName[_guide]}! Tap it to try it. This one is free!", 0); |
| SerpentClashGame.UI.cs:616 / void CompleteGuide(int b) | Toast("Nice! " + BoostName[b] + " is yours", HzUI.Hex("#FFE2A8"), 1.3f, IsPortrait ? 0.62f : 0.5f); |
| SerpentClashGame.UI.cs:704 / HzPopup TemplePopup(string title, string message, Color accent, Sprite art, string icon, params (string label, Color color, Action action)[] buttons) | var t = HzUI.Label("Title", ribbon.transform, title, 70, Color.white); t.rectTransform.Stretch(6); |

| | |
|---|---|
| `SerpentClashGame.UI.cs:710 / HzPopup TemplePopup(string title, string message, Color accent, Sprite art, string icon, params (string label, Color color, Action action)[] buttons)` | `p.Message = HzUI.Label("Msg", card.transform, message, 42, HzUI.Hex("#3D3020"), TextAnchor.MiddleCenter, false);` |
| `SerpentClashGame.UI.cs:753 / void UnlockPopup(int b, Action done)` | `TemplePopup("New: " + BoostName[b] + "!", BoostDesc[b] + "\nYou get " + BoostStart[b] + " free!", HzUI.Hex(BoostColor[b]), Art(BoostArt[b]), BoostIcon[b],` |
| `SerpentClashGame.UI.cs:760 / void ShowSettingsPopup()` | `var p = TemplePopup("Settings", "", Jade, null, null,` |
| `SerpentClashGame.UI.cs:761 / void ShowSettingsPopup()` | `(SoundLabel(), HzUI.Hex("#3E8ED0"), null),` |
| `SerpentClashGame.UI.cs:762 / void ShowSettingsPopup()` | `(MusicLabel(), HzUI.Hex("#8E6BD8"), null),` |
| `SerpentClashGame.UI.cs:773 / static string SoundLabel() => HzAudio.Muted ? "Sound: OFF" : "Sound: ON";` | `static string SoundLabel() => HzAudio.Muted ? "Sound: OFF" : "Sound: ON";` |
| `SerpentClashGame.UI.cs:774 / static string MusicLabel() => HzAudio.MusicMuted ? "Music: OFF" : "Music: ON";` | `static string MusicLabel() => HzAudio.MusicMuted ? "Music: OFF" : "Music: ON";` |

CHAPTER 11

# Artwork, fonts, audio and licences

Replace game images under Resources/serpentclash through Unity while retaining resource keys, .meta GUIDs, Sprite/UI import settings, alpha, pivots and slicing. Procedural graphics in the art/view sources may need separate colour changes. Replacing one PNG does not change every fallback. Keep third-party font and other licence notices with the corresponding shipped assets.

Effects are under Resources/serpentclash/Sfx. Preserve each actual load key and importer settings when replacing sounds. Supplied music: Resources/serpentclash/Music/temple_loop.ogg. Preserve that load key and verify loop/compression and the independent Music toggle. Kit UI uses supplied fonts; preserve actual licence notices and verify new glyphs.

CHAPTER 12

# Local saves, reset and migration

HzSave wraps PlayerPrefs and flushes local state on focus loss/quit. WebGL persistence uses browser storage tied to the hosting origin. Clearing site data or changing origin starts a new local save; this is not cloud synchronization. Inspect HzSave/PlayerPrefs calls in the actual game source before altering progression, stars, helper charges or tutorial flags.

Verify fresh launch, a completed arena, helper use, sound options, focus loss and restart. Use a dedicated test profile for resets. Keep technical save keys stable during colour/text reskins. If changing schema or keys, implement and test a migration rather than silently discarding existing users' progress.

CHAPTER 13

# Advertising integration and configuration

Asset: `Assets/Szekipapa77Ads/Resources/AdsSettings.asset`. Inspect backend, useGoogleTestIds and missingSdkFallback in the publisher-owned Ads configuration. Buyer projects include adapter/integration code and exclude external provider SDK binaries. WebGL and a missing-provider fallback simulate ads; these are not live commercial impressions.

For real Android ads install the supported provider SDK from its official source into your working project, enter your own app/placement IDs and use test ads first. Verify banner/interstitial/rewarded callbacks on supported hardware. For production use your own IDs and disable simulated missing-SDK fallback. Build tooling rejects unsafe production fallback settings. No IAP, remote backend or HLK is included in this publisher edition.

CHAPTER 14

# Android build and release

Install Unity 6000.0.62f1 Android modules through Hub. Select Android, run Setup and review application ID `com.szekipapa77.serpentclash`, version, scene orientation and signing. The build command writes `Builds/Android/serpentclash.apk`. Confirm BuildResult.Succeeded and the actual output file; intermediate report totals are not the final APK size.

Use your own keystore/signing and distribution settings. A test-signed demo is not a signed store release. A provider-free build simulates ads; a provider test-demo does not prove production ad-account configuration. Test real hardware touch, safe areas, focus/audio, save/relaunch and reward callbacks. Keep passwords, keystores and private service credentials out of buyer packages.

CHAPTER 15

# WebGL build and hosting

Install matching WebGL support. Run Setup and Build WebGL; output is `Builds/WebGL/serpentclash/`. Serve index.html with all loader/framework/data/WASM/StreamingAssets files over HTTP(S), not file://. Match compression, Content-Encoding and MIME types to the host. Outer page sizing and actual scene orientation are different settings.

The publisher's playable demo is hosted on HyperCodeStore. Upload the actual runtime files rather than merely changing a URL. Test a fresh anonymous load, real pointer controls, audio after interaction, saved progress/reload and phone/desktop sizing. Rebuild and rehost when code/assets change. A successful HTTP page alone is not gameplay verification.

CHAPTER 16

# Troubleshooting

**Compile errors:** restore matching packages and the full private Kit/Ads layer; only one game Kit belongs in a project. **Pink/invisible rendering:** run Setup and inspect Graphics and every Quality URP assignment. **No input:** classic input handling must be enabled; check canvas focus and blocking modals. **Missing resources:** restore exact paths, .meta GUIDs and importer settings.

**Edits do not appear:** stop Play, Save in the level editor, reload the canonical bank and check current saved level. **Text reverts:** update construction and refresh assignments from the exact table. **No live ads:** provider binaries are deliberately excluded; inspect your installed provider and fallback mode. **Scene reload exception:** retain owner-lifetime and generation guards on delayed callbacks. **WebGL fails:** inspect browser Console, compression headers and missing payloads.

CHAPTER 17

# Support and final release checklist

Publisher: Szekipapa77. Support: szekipapa77@gmail.com. Include game title, exact Unity version, platform, first Console error and reproduction steps. Never send passwords/tokens.

1. Clean-import the standalone project and native package.
2. Play both scenes and replace a scene near startup.
3. Review tutorial, normal completion/retry and every helper activation plus effect.
4. Validate editor changes and test reskin text/colours in both layouts.
5. Rebuild affected Android/WebGL outputs.
6. Verify complete public APK bytes, actual browser demo, documentation and video links.
7. Keep licences and exclude all internal prompts, planning/capture/build/QA tooling from buyers.